

## Markov Matrix:

$$A = \begin{bmatrix} 0.1 & 0.01 & 0.3 \\ 0.2 & 0.99 & 0.3 \\ 0.7 & 0 & 0.4 \end{bmatrix}$$

①  $A^k \geq 0$

② Sum of Cols<sup>k</sup> = 1

Steady state for Powers

$$\lambda = 1$$

1.  $\lambda = 1$  is eigen-value

2. All other  $\lambda$ s are  $< 1$

$$u_k = A^k u_0$$

$$= c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 + \dots$$

$\rightarrow 0$  as  $t \rightarrow \infty$

$$= c_1 x_1$$

$\rightarrow$  part of  $u_0$

$$A - 1I = \begin{bmatrix} -0.9 & 0.01 & 0.3 \\ 0.2 & -0.01 & 0.3 \\ 0.7 & 0 & -0.6 \end{bmatrix}$$

$\hookrightarrow$  Cols  $(A - I)$  add to 0

$\hookrightarrow (A - I)$  is singular

Application:

$$u_{k+1} = A u_k$$

$$\begin{bmatrix} u_{cal} \\ u_{mass} \end{bmatrix} = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix} \begin{bmatrix} u_{cal} \\ u_{mass} \end{bmatrix}$$

$\hookrightarrow$  Markov matrix

$$\begin{bmatrix} u_{cal} \\ u_{mass} \end{bmatrix}_0 = \begin{bmatrix} 0 \\ 1000 \end{bmatrix}$$

$$\lambda = 1, 0.7$$

$$\lambda_1 = 1$$

$$\begin{bmatrix} -0.1 & 0.2 \\ 0.1 & -0.2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\hookrightarrow x_1$

$$\lambda_2 = 0.7$$

$$\begin{bmatrix} 0.2 & 0.2 \\ 0.1 & 0.1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\hookrightarrow x_2$

Now,

$$a_k = c_1 1^k \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 (0.7)^k \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

At  $k=0$ ,

$$\begin{bmatrix} 0 \\ 1000 \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$c_1 = \frac{1000}{3}, \quad c_2 = \frac{2000}{3}$$

Projections with orthonormal basis  $\{b_1, \dots, b_n\}$

$$V = x_1 b_1 + x_2 b_2 + \dots + x_n b_n$$

$$\begin{bmatrix} b_1 & \dots & b_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = V$$

$$Qx = V$$

$$x = Q^{-1}V$$

$$= Q^T V$$

Fourier Series:

$$f(x) = a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \dots \quad \nearrow \text{Infinite}$$

$$B = \{1, \cos x, \sin x, \cos 2x, \sin 2x, \dots\} \rightarrow \text{orthogonal}$$

Dot product for functions:

$$f^T g = \int_0^{2\pi} f(x) g(x) dx$$

$$v^T w = \sum_{i=1}^n v_i w_i$$

$$a_1 = \frac{\int_0^{2\pi} f(x) \cos x dx}{\int_0^{2\pi} (\cos x)^2 dx}$$